# evaluate_jsdm.R

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`evaluate_jsdm` *Evaluate JSDM Model Performance*

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### Description

Evaluates a fitted sjSDM model and returns comprehensive performance metrics at both model and species level.

### Usage

```
evaluate_jsdm(mod_jsdm = NULL)
```

### Arguments

| | |
|---|---|
| `mod_jsdm` | A fitted sjSDM model object. Must be of class 'sjSDM'. |

### Details

For binomial models, species-level classification metrics (AUC, Accuracy, LogLoss) are computed using a 0.5 probability threshold for binary predictions. For other model families, RMSE is computed per species.

Convergence is assessed via [diagnose_jsdm_convergence()], which analyses the training loss history. A 'linear_trend_slope' < 0.01 and 'median_diff' < 1 in the returned 'convergence' element indicate that the model has converged.

### Value

A list with three elements: - 'vec_model_metrics': Named numeric vector of R-squared values (McFadden, Nagelkerke) - 'data_taxon_metrics': A tibble with one row per taxon and columns: 'taxon_name', 'auc', 'accuracy', 'log_loss' (binomial) or 'rmse' (other families) - 'list_convergence_diagnostics': A list from [diagnose_jsdm_convergence()] with 'linear_trend_slope', 'median_diff', 'convergence_plot', and 'note'

### See Also

sjSDM::Rsquared, Metrics::auc, diagnose_jsdm_convergence

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